

ONE PAGE EXECUTIVE SUMMARY

Customer Behavior Analysis

Alfido Tech Data Analytics Internship

Project Overview

This project was conducted as part of the Alfido Tech Data Analytics Internship to analyze customer purchasing behavior using real-world e-commerce transaction data. The primary objective was to clean and preprocess the dataset, perform exploratory data analysis (EDA), identify customer segments through RFM analysis and K-Means clustering, analyze purchase patterns and retention trends, and generate actionable business recommendations.

Two customer datasets were combined and processed using Python libraries including Pandas, NumPy, Matplotlib, Seaborn, and Scikit-learn. Data preprocessing involved removing duplicate records, validating data types, converting date fields, handling missing values, and creating new analytical features such as Month, Year, Day, Age Group, Purchase Level, and Customer Lifetime Value (CLV).

Exploratory Data Analysis (EDA) was performed to understand customer demographics, product preferences, payment methods, purchase amounts, seasonal trends, and customer behavior. Multiple visualizations were created to support business interpretation and identify meaningful patterns.

Customer segmentation was carried out using the RFM (Recency, Frequency, Monetary) model followed by K-Means clustering. Customers were grouped into four meaningful business segments: **High Value Customers**, **Loyal Customers**, **Regular Customers**, and **At Risk Customers**. These segments provide valuable insights for designing targeted marketing strategies and improving customer engagement.

Purchase pattern analysis revealed seasonal variations in sales, while retention analysis highlighted the importance of maintaining continuous customer engagement. Customer Lifetime Value (CLV) analysis identified the most profitable customers, enabling businesses to prioritize high-value relationships.

Three Actionable Insights

1. Strengthen Customer Loyalty Programs

High-value and loyal customers contribute significantly to business revenue. Reward-based loyalty programs, exclusive offers, and premium membership benefits can improve retention and increase customer lifetime value.

2. Re-engage At-Risk Customers

Customers identified as at risk should receive personalized promotional offers, reminder emails, and targeted marketing campaigns to reduce churn and encourage repeat purchases.

3. Implement Data-Driven Personalized Marketing

Customer segmentation should be integrated into marketing strategies. Personalized product recommendations and behavior-based campaigns can improve customer satisfaction, increase conversion rates, and optimize marketing investment.

Conclusion

This project demonstrates how data analytics techniques such as data cleaning, exploratory analysis, RFM segmentation, K-Means clustering, and business intelligence can transform raw customer data into actionable insights. The results support informed decision-making and provide practical recommendations for improving customer retention, increasing revenue, and enhancing overall business performance.